SPSS Tutorial for Research Data Analysis

Introduction to SPSS

SPSS (Statistical Package for the Social Sciences) is powerful software for statistical analysis commonly used in social science, business, and health research.

GETTING STARTED

Opening SPSS

1. Launch SPSS Statistics
2. Choose "New Dataset" or "Open an existing data source"
3. Familiarize yourself with two main views:
   - **Data View**: Spreadsheet for entering data
   - **Variable View**: Define variable properties

DATA ENTRY AND SETUP

Variable View Setup

For each variable, define:

Property	Description	Example
Name	Variable identifier (no spaces)	age, gender, test_score
Type	Numeric, String, Date	Numeric
Width	Number of characters	8
Decimals	Decimal places	2
Label	Descriptive name	"Age in years"
Values	Code meanings	1=Male, 2=Female
Missing	Missing data codes	999
Measure	Scale/Ordinal/Nominal	Scale for continuous data

Example Variable Setup

```
Name: gender
```

```
Type: Numeric
Label: "Participant Gender"
Values: 1 = "Male", 2 = "Female"
Measure: Nominal

Name: exam_score
Type: Numeric
Decimals: 1
Label: "Final Exam Score"
Measure: Scale
```

BASIC STATISTICAL ANALYSES

1. DESCRIPTIVE STATISTICS

Purpose: Summarize data characteristics

Steps:

1. **Analyze → Descriptive Statistics → Descriptives**
2. Select variables to analyze
3. Click "Options" to choose statistics:
   - Mean, Median, Mode
   - Standard Deviation, Variance
   - Minimum, Maximum, Range
4. Click OK

Output Interpretation:

```
Descriptive Statistics
            N   Minimum  Maximum  Mean    Std. Deviation
Exam_Score    80  45.0     98.0     76.85   12.34
Age         80  18      28      21.30   2.10
```

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2. FREQUENCY TABLES

Purpose: Count occurrences of each value

Steps:

1. **Analyze → Descriptive Statistics → Frequencies**
2. Select categorical variables (gender, education level)
3. Click "Statistics" for additional measures
4. Click "Charts" for visual representations
5. Click OK

Use For:

- Categorical data (gender, ethnicity)
- Checking data entry errors
- Understanding distribution

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3. T-TESTS

Independent Samples T-Test

Purpose: Compare means of two independent groups

Example: Do males and females have different exam scores?

Steps:

1. **Analyze → Compare Means → Independent-Samples T Test**
2. Move dependent variable (exam_score) to "Test Variable"
3. Move grouping variable (gender) to "Grouping Variable"
4. Click "Define Groups": Group 1 = 1, Group 2 = 2
5. Click OK

Output Interpretation:

```
Levene's Test: p > .05 → Equal variances assumed
t-test: t(78) = 2.45, p = .017

Interpretation: Males (M = 79.2) scored significantly higher
than females (M = 74.5), t(78) = 2.45, p < .05.
```

Paired Samples T-Test

Purpose: Compare means of same group at two time points

Example: Did scores improve from pre-test to post-test?

Steps:

1. **Analyze → Compare Means → Paired-Samples T Test**
2. Select both variables (pre_test, post_test)
3. Move pair to "Paired Variables"
4. Click OK

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4. ONE-WAY ANOVA

Purpose: Compare means across three or more groups

Example: Do exam scores differ across three teaching methods?

Steps:

1. **Analyze → Compare Means → One-Way ANOVA**
2. Dependent Variable: exam_score
3. Factor: teaching_method
4. Click "Post Hoc": Select Tukey or Bonferroni
5. Click "Options": Check Descriptive statistics
6. Click OK

Output Interpretation:

```
ANOVA
            df   F      Sig.
Between Groups  2   8.45   .001

Post Hoc Tests (Tukey):
Method A vs Method B: p = .023 (significant)
Method A vs Method C: p = .001 (significant)
Method B vs Method C: p = .156 (not significant)
```

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5. CORRELATION ANALYSIS

Purpose: Measure strength and direction of relationship between variables

Steps:

1. **Analyze → Correlate → Bivariate**
2. Select variables to correlate
3. Choose correlation coefficient:
   - **Pearson**: For continuous variables
   - **Spearman**: For ordinal or non-normal data
4. Check "Two-tailed" test
5. Click OK

Output Interpretation:

```
Correlations
              Study_Hours  Exam_Score
Study_Hours     1           .652**
Exam_Score      .652**       1

** Correlation is significant at p < .01

Interpretation: Strong positive correlation (r = .652, p < .01)
More study hours associated with higher exam scores.
```

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6. REGRESSION ANALYSIS

Purpose: Predict dependent variable from independent variable(s)

Steps:

1. **Analyze → Regression → Linear**
2. Dependent: exam_score
3. Independent(s): study_hours, attendance_rate
4. Click "Statistics": Check Estimates, Model fit, Descriptives
5. Click OK

Output Interpretation:

```
Model Summary
R = .712, R² = .507, Adjusted R² = .494

ANOVA: F(2,77) = 39.45, p < .001 (Model is significant)

Coefficients:
                B     SE    Beta   t      p
(Constant)      32.5  5.2          6.25   .000
Study_Hours      3.2  0.6   .445   5.33   .000
Attendance_Rate  0.8  0.3   .223   2.67   .009

Interpretation: Model explains 50.7% of variance.
Each additional study hour predicts 3.2 point increase in score.
```

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7. CHI-SQUARE TEST

Purpose: Test association between categorical variables

Example: Is gender associated with pass/fail status?

Steps:

1. **Analyze → Descriptive Statistics → Crosstabs**
2. Row: gender
3. Column: pass_fail
4. Click "Statistics": Check Chi-square
5. Click "Cells": Check Expected, Row percentages
6. Click OK

Output Interpretation:

```
Chi-Square Tests
Pearson Chi-Square: χ²(1) = 6.42, p = .011

Interpretation: Significant association between gender and
pass/fail status, χ²(1) = 6.42, p < .05.
```

DATA VISUALIZATION

Creating Bar Charts

Graphs → Legacy Dialogs → Bar

- Choose Simple, Clustered, or Stacked
- Define categories and values
- Add titles and labels

Creating Histograms

Graphs → Legacy Dialogs → Histogram

- Select variable
- Check "Display normal curve" to assess normality

Creating Scatterplots

Graphs → Legacy Dialogs → Scatter/Dot

- Select Simple Scatter
- Define X and Y variables
- Add fit line for trend

DATA MANAGEMENT TIPS

Recoding Variables

Transform → Recode into Different Variables

- Create categories from continuous data
- Example: Age → Age groups (18-20, 21-23, 24+)

Computing New Variables

Transform → Compute Variable

- Create composite scores
- Example: Total_Score = Test1 + Test2 + Test3

Handling Missing Data

Transform → Replace Missing Values

- Mean substitution
- Regression imputation
- Delete cases with missing data (listwise/pairwise)

Selecting Cases

Data → Select Cases

- Analyze subset of data
- Example: Only analyze female participants

COMMON ERRORS AND SOLUTIONS

Error: "Variables in the formula are not all numeric"

Solution: Check variable types in Variable View; convert strings to numeric if needed.

Error: "Output appears empty"

Solution: Ensure you selected variables and clicked OK; check if cases meet selection criteria.

Warning: "Levene's test is significant"

Solution: Use "Equal variances not assumed" row in t-test output.

Issue: "ANOVA assumes normality"

Solution: Check normality with histograms/Q-Q plots; consider non-parametric alternatives.

REPORTING RESULTS

General Format

"A [test name] was conducted to [purpose]. Results indicated [significant/non-significant] [direction], [test statistic], p [= or < value]."

Examples

T-Test: "An independent samples t-test revealed that the experimental group (M = 85.3, SD = 7.2) scored

significantly higher than the control group ($M = 78.6$, $SD = 8.1$), $t(78) = 3.67$, $p < .001$."

ANOVA: "A one-way ANOVA showed significant differences in exam scores across the three teaching methods, $F(2, 117) = 12.45$, $p < .001$, $\eta^2 = .18$."

Correlation: "Study hours were positively correlated with exam performance, $r(78) = .65$, $p < .001$."

Regression: "Multiple regression analysis revealed that study hours ($\beta = .45$, $p < .001$) and attendance ($\beta = .22$, $p < .01$) significantly predicted exam scores, $R^2 = .51$, $F(2, 77) = 39.45$, $p < .001$."

CHECKLIST FOR SPSS ANALYSIS

Before Analysis:

- [ ] Data entered correctly in Data View
- [ ] Variables defined properly in Variable View
- [ ] Missing data coded consistently
- [ ] No data entry errors (check frequencies)
- [ ] Variables are correct type (numeric for calculations)

During Analysis:

- [ ] Appropriate test selected for research question
- [ ] Assumptions checked (normality, homogeneity)
- [ ] Correct variables selected
- [ ] Options and statistics chosen appropriately

After Analysis:

- [ ] Output saved with descriptive filename
- [ ] Results interpreted correctly
- [ ] Effect sizes calculated when appropriate
- [ ] Graphs created for visualization
- [ ] Findings documented for report

ADDITIONAL RESOURCES

SPSS Help

- **Help → Topics**: Built-in tutorials
- **Help → Tutorial**: Step-by-step guides

- IBM SPSS website: Official documentation

Online Resources

- YouTube tutorials for specific analyses
- University statistics support centers
- SPSS community forums

Books

- Field, A. (2018). *Discovering Statistics Using IBM SPSS Statistics*
- Pallant, J. (2020). *SPSS Survival Manual*

PRACTICE EXERCISES

Exercise 1: Descriptive Statistics

Import sample dataset, calculate means and standard deviations for all continuous variables.

Exercise 2: T-Test

Compare two groups on an outcome variable; report results in APA format.

Exercise 3: Correlation

Examine relationships between three continuous variables; create correlation matrix.

Exercise 4: Regression

Predict an outcome from multiple predictors; interpret coefficients and R^2.

This tutorial covers essential SPSS operations for research students. Practice with your own data to build proficiency.

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